100/100

Signed off

Week 4 complete, Timothy

The notebook meets the rubric fully, with correct code and well-justified written answers across all tasks.

Task 1 Bins by hand — 20/20

The code uses both `pd.cut(ages, bins=3, retbins=True)` and `pd.qcut(ages, q=4, retbins=True)`. The written answer correctly gives the equal-width calculation, the expected counts, and the outlier/90-year-old reflection.

Task 2 Age two ways — 25/25

The notebook creates `age_ew` and `age_ef` with `retbins=True`, prints edges and ordered counts, computes deterioration rates with `groupby(..., observed=True).agg(['count','mean'])`, and plots two side-by-side bar charts with shared y-axis. The written answer chooses equal frequency for the ward manager and justifies it with the small first equal-width bin and the more stable patient counts.

Task 3 Unseen patients — 25/25

The code splits with `train_test_split(test_size=0.25, random_state=42)`, fills heart rate using the training median, learns six equal-frequency edges on train, applies them to test, bins the new patients, and then opens the ends with `-np.inf` and `np.inf`. The written answer explains that closed edges make 190 bpm fall outside the range and that open ends fix the issue.

Task 4 Choose the method — 15/15

All four columns are addressed with sensible binning choices and a clear statement of what is lost in each case. The answers are consistent with the rubric's acceptable alternatives, including equal frequency for skewed income and visit gaps, equal width for exam marks and body temperature.

Reflection — 15/15

The reflection gives more than three sentences, explains when equal width is preferable versus equal frequency, and cites specific notebook results, including the age counts 20, 172, 494, 383, 131 and the sparse first bin. It clearly links the evidence to the choice of method.

Reference	**SO-8TQZ-T38A**
Submitted	20 October 2026, 15:07 UTC
Attempt	1 of 2

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